Processos Estocásticos (Dia 19/10/2026)

Teo: Dado $t \in [0,1]$ fixado

$$(\heartsuit) \quad \sum_{n=1}^{N} Z_n \langle 1_{[0,t]}, \phi_n \rangle$$

converge em L^2 e q.c quando $N \to \infty$. X_t tá bem definido

e $(X_t)_{t \geq 0} \sim GP(0, K)$, $K(s,t) = s \wedge t$.

Prova: Defina $M_N^{(t)} = \sum_{K=1}^{N} Z_K \langle 1_{[0,t]}, \varphi_K \rangle$

$(M_N^{(t)})_{N \in \mathbb{N}}$ é martingal

- $\mathbb{E}\left((M_n^{(t)})^2\right) = \sum_{K=1}^{N} \underbrace{\mathbb{E}(Z_K^2)}_{=1} \langle 1_{[0,t]}, \varphi_K \rangle^2$

$$\leq \sum_{K=1}^{\infty} \quad \text{——————}$$

$$= \|1_{[0,t]}\|^2 = t.$$

Teo. Convergência de Martingais em $L^2 \Rightarrow M_n^{(t)} \to X_t$ q.c e em L^2.

Convergência $L^2 \Rightarrow$
$$\begin{cases} \mathbb{E}(X_t) = \lim_{N \to \infty} \mathbb{E}(M_N^{(t)}) \\ Cov(X_t, X_s) = \lim_{N \to \infty} Cov(M_N^{(t)}, M_N^{(s)}) \end{cases}$$

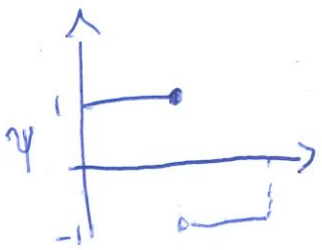
$\Rightarrow \mathbb{E}(X_t) = 0$

$Cov(X_s, X_t) = s \wedge t.$

$(M_N^{(t)})_{t = t_1, \dots, t_m}$ é gaussiano $\Rightarrow (X_t)_{t = t_1, \dots, t_m}$ é gaussiano. ∎

~~(1)~~ (2)

• Seja $\psi(t) = \mathbb{1}_{[0,1/2]}(t) - \mathbb{1}_{(1/2,1]}(t)$, $\forall\, t \in [0,1]$.



• $\psi_0 \equiv 1$

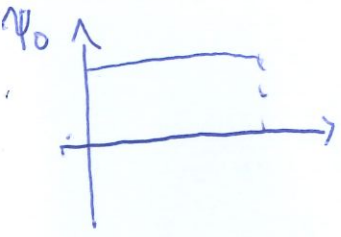
• $\psi_n(t) = 2^{j/2}\psi(2^j t + K)$ $\forall n = 2^j + K$

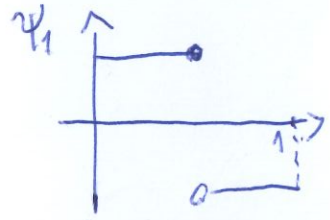
$j \in \mathbb{N}$, $K = 0, \dots, 2^j - 1$.

$\Rightarrow$ $(\psi_n)_{n\geq 0}$ é base ortonormal.

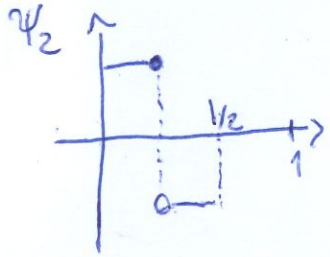
Seja $\Delta_n(t) = \int_0^t \psi_n(s)\,ds$.

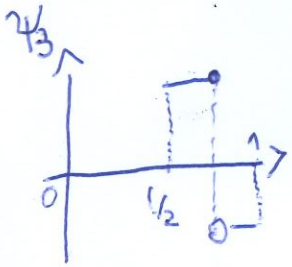
$\Rightarrow$ Δ_n é contínua, $\|\Delta_n\|_\infty = 2^{-j/2-1}$ $\forall j \geq 0$



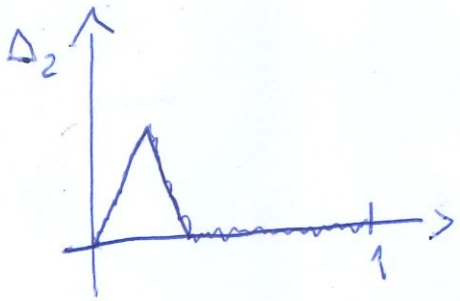


$j = 0$





$j = 1$



<u>Lema</u>: Seja $(Z_n)_{n\in\mathbb{N}}$ iid $N(0,1)$ $\Rightarrow \exists$ v.a. C, $\mathbb{P}(C<\infty) = 1$ t.q.

(M) $|Z_n| \leq C\sqrt{\log n}$, $\forall n \geq 2$.

<u>prova</u>: $\forall x \geq 1$,

$$\mathbb{P}(|Z_n| \geq x) = \sqrt{\frac{2}{\pi}} \int_x^\infty e^{-y^2/2} \leq \sqrt{\frac{2}{\pi}} \int_x^\infty y e^{-y^2/2}\,dy = \sqrt{\frac{2}{\pi}}\, e^{-x^2/2}$$

(3)

$$\Rightarrow \mathbb{P}\left(|Z_n| \geq \sqrt{2\alpha \log n}\right) \leq \sqrt{\frac{2}{\pi}}\, e^{-\alpha \log n} = \sqrt{\frac{2}{\pi}}\, n^{-\alpha}$$

Borel Cantelli $\Rightarrow \forall \alpha > 1,\ \mathbb{P}\left(|Z_n| \geq \sqrt{2\alpha \log n} \text{ i.o.}\right) = 0.$

$$C := \sup_{n \geq 2} \frac{|Z_n|}{\sqrt{\log n}} \Rightarrow \mathbb{P}(C = \infty) = 0. \quad \blacksquare$$

Teo: Seja $\Delta_n(t) = \langle \mathbb{1}_{[0,t]}, \psi_n \rangle$. Seja

$$M_N(t) = \sum_{n=0}^{\infty} Z_n \Delta_n(t) \quad \forall t \in [0,1]$$

$\Rightarrow M_N \in L^\infty([0,1])$, e $\exists M_\infty \in L^\infty([0,1])$ aleatório, t.q $\quad (*)$

$\|M_N - M_\infty\|_\infty \xrightarrow[N \to \infty]{} 0$ q.c.

Além disso, M_∞ é um movimento Browniano em $[0,1]$.

prova: Dado o teorema anterior (♡), basta provar $(*)$.

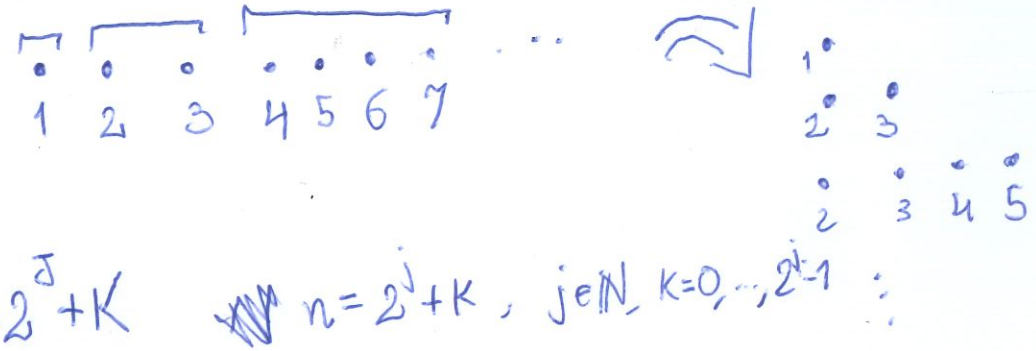
Pelo lema (N), $\exists$ C aleatória t.q

$$\sum_{n=N} |Z_n| \Delta_n(t) \leq C \sum_{n=N} \sqrt{\log n}\, \Delta_n(t) \quad \forall t \in [0,1]$$

$$\sum_{n=N}^{\infty} |Z_n| \Delta_n(t) \leq C \sum_{n=N}^{\infty} \sqrt{\log n}\, \Delta_n(t) \quad \forall t \in [0,1],\ \forall N$$



(4)



$N = 2^J + K$ $\quad n = 2^j + K$, $j \in \mathbb{N}$, $K = 0, \dots, 2^j - 1$:

$$\sum_{n=N}^{\infty} \sqrt{\log n}\, \Delta_n(t) = \sum_{j=J}^{\infty} \sum_{K=0}^{2^j-1} \sqrt{\log(2^j+K)}\, \Delta_{2^j+K}(t)$$

$$\leq \sum_{j=J}^{\infty} \sqrt{\log(2^j+K)}\, 2^{-j/2-1} \leq \sum_{j=J}^{\infty} \sqrt{j+1}\, 2^{-j/2-1} \longrightarrow 0$$

$N \to \infty$ ∎

Propriedades da trajetória

Def: $\varphi : [0,1] \longrightarrow \mathbb{R}$ é α-Hölder contínua, $\alpha > 0$, se existe $c > 0$ t.q $|\varphi(t) - \varphi(s)| \leq c|t-s|^{\alpha}$ $\forall t, s \in [0,1]$.

Teo: $t \longmapsto B_t(\omega)$ é α-Holder contínuo w.q.c, $\forall \alpha \in (0, 1/2)$.

Prova:

- $D_j(t,s) := \sum_{n=2^j}^{2^{j+1}-1} Z_n \left(\Delta_n(t) - \Delta_n(s) \right)$

$$\Rightarrow B_t - B_s = \sum_{j=0}^{\infty} D_j(t,s)$$

(5) $n = 2^j + k$, $k = 0, \dots, 2^j - 1$

$\Rightarrow |Z_n| \le C\sqrt{\log(2^j+k)} \le C\sqrt{j+1}$

- $|\Delta_n(t) - \Delta_n(s)| \le 2\,\|\Delta\|_\infty \le 2 \cdot 2^{-j/2-1} = 2^{-j/2}$
- $|\Delta_n(t) - \Delta_n(s)| \le \int_s^t |\psi_n(u)|\,du \le |t-s| \cdot 2^{j/2}$

$\Rightarrow |\Delta_n(t) - \Delta_n(s)| \le 2^{-j/2} \wedge (|t-s|2^{j/2})$

$\exists! k \in \{0, \dots, 2^j - 1\}$ t.q. $\Delta_n(t) \neq 0$.

$\exists! k \in \{0, \dots, 2^j - 1\}$ t.q. $\Delta_n(s) \neq 0$.

$$\Rightarrow |D_j(t,s)| \le C\sqrt{j+1} \sum_{n=2^j}^{2^{j+1}-1} |\Delta_n(t) - \Delta_n(s)|$$

$$\le C\sqrt{j+1} \cdot 2 \cdot \max_{n=2^j,\dots,2^{j+1}-1} |\Delta_n(t) - \Delta_n(s)|$$

$$\le C\sqrt{j+1} \cdot 2 \cdot \left(2^{-j/2} \wedge (|t-s|\,2^{j/2})\right)$$

- $\forall \alpha < 1/2$, $\sqrt{j+1}\, 2^{-j/2+\alpha j} \to 0$. Logo, $\exists \tilde{c} > 0$ t.q.

$$\sqrt{j+1}\, 2^{-j/2} \le \tilde{c}\, 2^{-\alpha j} \quad \forall j \ge 0$$

$$\Rightarrow |D_j(t,s)| \le \tilde{c}\, 2^{-\alpha j}\left(1 \wedge (2^j|t-s|)\right).$$

(6)

$$|B_t - B_s| \leq \sum_{j=0}^{\infty} |D_j(t,s)|$$

$$= \sum_{j=0}^{K} |D_j(t,s)| + \sum_{j=K+1}^{\infty} |D_j(t,s)| \qquad \text{(pra qualquer } K>0\text{)}$$

$$\leq \tilde{C}\left(\sum_{j=0}^{K} 2^{-\alpha j} (1 \wedge (2^j |t-s|)) + \sum_{j=K+1}^{\infty} 2^{-\alpha j} (1 \wedge (2^j |t-s|)) \right)$$

$$\leq \tilde{C}\left(\sum_{j=0}^{K} 2^{-\alpha j} \cdot 2^j |t-s| + \sum_{j=K+1}^{\infty} 2^{-\alpha j} \right)$$

Escolha $K \in \mathbb{N}$ t.q $2^{-K-1} \leq |t-s| \leq 2^{-K}$.

$$\leq \tilde{C}\left(|t-s| \frac{2^{(1-\alpha)(K+1)} - 1}{2^{1-\alpha} - 1} + \frac{2^{-\alpha(K+1)}}{1 - 2^{-\alpha}} \right)$$

Escolha $k \in \mathbb{N}$ t.q. $2^{-K-1} \leq |t-s| \leq 2^{-K}$.

$$\Rightarrow |t-s| \frac{2^{(1-\alpha)(K+1)} - 1}{2^{1-\alpha} - 1} + \frac{2^{-\alpha(K+1)}}{1-2^{-\alpha}} \leq 2^{-K} \frac{2^{(1-\alpha)(K+1)}}{2^{1-\alpha} - 1} + \frac{2^{-\alpha(K+1)}}{1-2^{-\alpha}}$$

$$= \frac{2}{2^{1-\alpha} - 1} 2^{-\alpha(K+1)} + \frac{1}{1-2^{-\alpha}} 2^{-\alpha(K+1)} \leq c(\alpha) |t-s|^{\alpha}$$